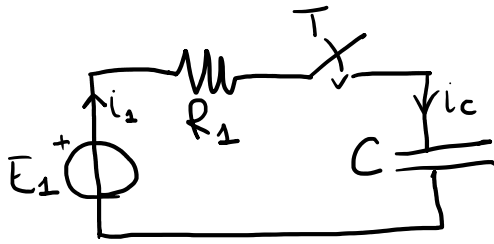


Esercizio

venerdì 8 aprile 2022 09:12



$$R_1 = 100 \, \Omega$$

$$C_1 = 100 \, \mu\text{F}$$

$$E_1 = 10 \, \text{V}$$

T si chiude $t=0$

Cond. è inizialmente scarica

$$1) P_{E_1, \max} = ? \quad t_{P_{E_1, \max}} = ?$$

$$2) P_{E_1}(t=\infty) = ?$$

$$3) t_{P_E^*} = ? \quad P_E^* = \frac{1}{2} P_{E_1, \max}$$

$$P_{E_1} = E_1 \cdot i_1$$

$$t=0^- : v_C(t=0^-) = 0 \, \text{V}$$

$$t=0^+ : v_C(t=0^+) = v_C(t=0^-) = 0 \, \text{V}$$

$$t=\infty : v_C(t=\infty) = E_1 = 10 \, \text{V}$$

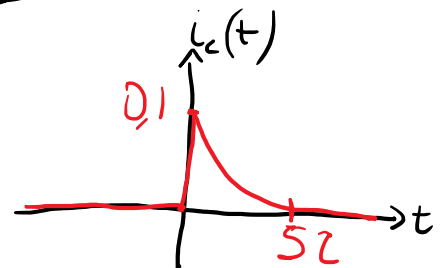
$$x(t) = (X_0 - X_\infty) e^{-t/\tau} + X_\infty$$

$$v_C(t) = (0 - 10) e^{-t/\tau} + 10 \, \text{V} = 10(1 - e^{-t/\tau})$$

$$\tau = R_1 C = 10 \, \text{ms}$$

$$i_C = C \frac{dv_C}{dt} = C \frac{d}{dt} [10(1 - e^{-t/\tau})] = +C 10 \left(+\frac{1}{\tau}\right) e^{-t/\tau} =$$

$$= \frac{10}{R} e^{-t/\tau} = 0,1 e^{-t/\tau} \, [\text{A}]$$



$$1) P_{E_1, \max} = E_1 \cdot I_{1, \max} = 10 \cdot 0,1 = 1 \text{ W}$$

$$t_{P_{E_1, \max}} = 0 \text{ s}$$

$$2) P_{E_1}(t=\infty) = E_1 \cdot I_1(t=\infty) = 0 \text{ W}$$

$$3) P_{S_1} = E_1 \cdot i_C = E_1 \cdot 0,1 e^{-t/\tau} \rightarrow E_1 \cdot 0,1 e^{-\frac{t_{P_{S_1}}}{\tau}} = \frac{P_{S_1, \max}}{2} = 0,5$$

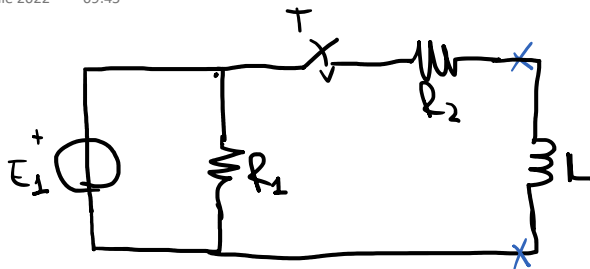
$$e^{-\frac{t_{P_{S_1}}}{\tau}} = 0,5 \rightarrow -\frac{t_{P_{S_1}}}{RC} = \ln(0,5)$$

$$\rightarrow t_{P_{S_1}} = -RC \ln(0,5) = +RC(+0,69)$$

$$= 0,01(0,69) = 6,9 \text{ ms}$$

Esercizio 2

venerdì 8 aprile 2022 09:45



$$E_1 = 12 \text{ V}$$

$$R_1 = R_2 = 10 \Omega$$

$$L = 0,5 \text{ H}$$

Ind. iniziale scarica

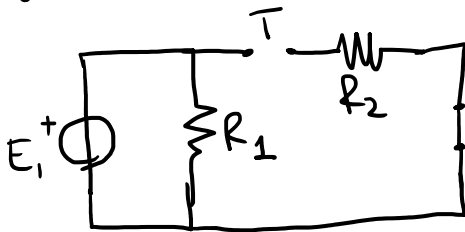
T si chiude in $t=0$

1) $I_{E_1}(t=0^-) = ?$

2) $I_{E_2}(t=\infty) = ?$

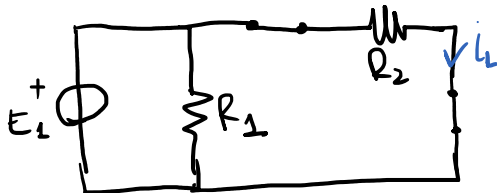
3) espressione analitica di P_{E_1}

① $t=0^-$



$$\rightarrow I_{E_1}(t=0^-) = \frac{E_1}{R_1} = 1,2 \text{ A}$$

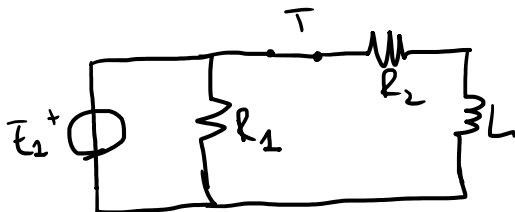
② $t=\infty$



$$\rightarrow I_{E_2}(t=\infty) = \frac{E_1}{R_1 // R_2} = \frac{E_1}{5 \Omega} = 2,4 \text{ A}$$

$$I_L(t=\infty) = \frac{E_1}{R_2} = 1,2 \text{ A}$$

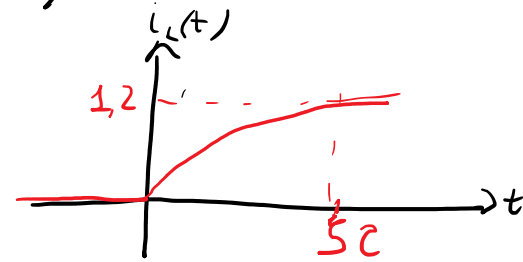
③



$$P_{E_1} = E_1 \cdot i_1 = E_1 [i_{R_2} + i_L]$$

$$i_L(t) = (I_{L0} - I_{L\infty}) e^{-t/\tau} + I_{L\infty} = (0 - 1,2) e^{-t/2} + 1,2 \text{ A}$$

$$\tau = \frac{L}{R} = \frac{0,5 \text{ H}}{10 \text{ }\Omega} = 50 \text{ ms}$$



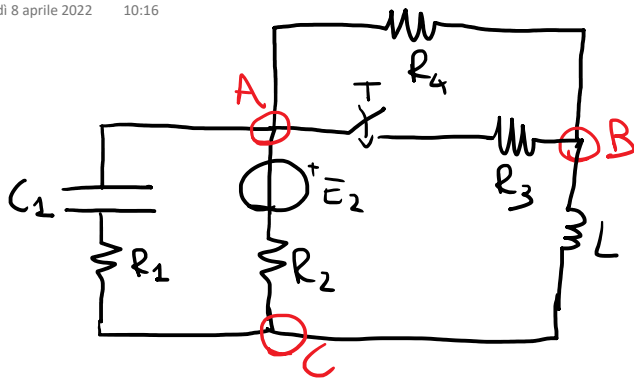
$$P_{E1} = E_1 \left[\frac{E_1}{R_1} + 1,2(1 - e^{-t/\tau}) \right]$$

$$= 12 \left[1,2 + 1,2 - 1,2 e^{-t/\tau} \right] = 12 \left[2,4 - 1,2 e^{-t/\tau} \right] =$$

$$= 28,8 - 14,4 e^{-t/\tau} \text{ W}$$

Esercizio 3

venerdì 8 aprile 2022 10:16



$$E_2 = 12 \text{ V}$$

$$R_1 = R_2 = 2 \text{ } \Omega$$

$$R_3 = R_4 = 4 \text{ } \Omega$$

$$C_1 = 10 \text{ } \mu\text{F} \quad L_1 = 2 \text{ mH}$$

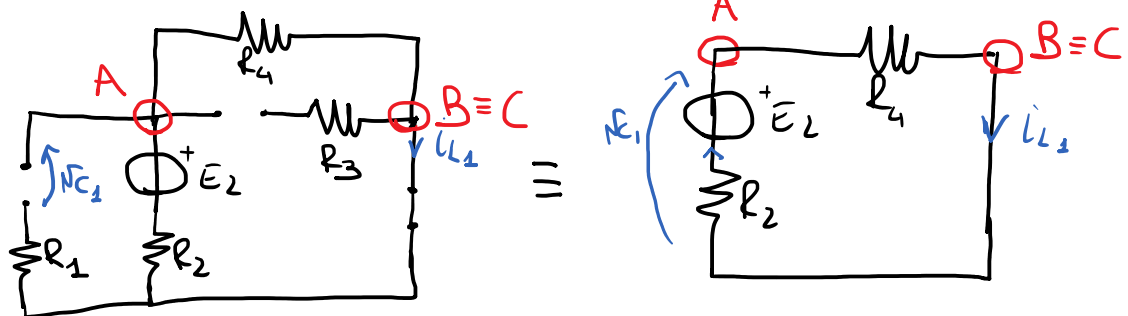
T si chiude in $t=0$

$$t=0^-: \quad Q_{C_1} = ?$$

$$t=0^+: \quad i_{C_1} = ? ; \quad \frac{di_L}{dt} = ?$$

$$t=\infty: \quad \mathcal{E}_{L_1} = ? ; \quad \mathcal{E}_{C_1} = ?$$

$t=0^-$ $\rightarrow$ regime stazionario $\Rightarrow \frac{d}{dt} = 0$



$$i_{L_1}(t=0^-) = \frac{E_2}{R_2 + R_4} = 2 \text{ A}$$

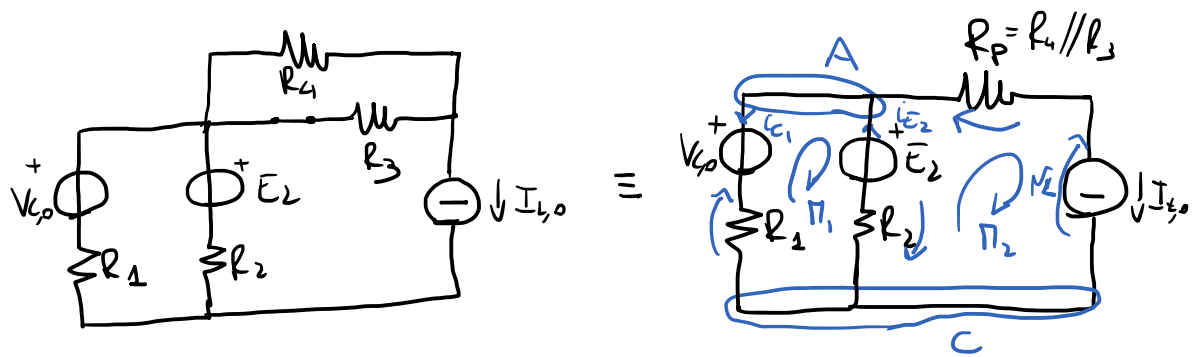
$$U_{C_1}(t=0^-) = R_4 \cdot i_{L_1} = 8 \text{ V}$$

$$Q_{C_1}(t=0^-) = C_1 \cdot U_{C_1}(t=0^-) = 10 \text{ } \mu\text{F} \cdot 8 \text{ V} = 8 \cdot 10^{-5} \text{ C}$$

$t=0^+$

$$i_{L_1}(t=0^+) = i_{L_1}(t=0^-) = I_{L_0} = 2 \text{ A}$$

$$U_{C_1}(t=0^+) = U_{C_1}(t=0^-) = V_{C_0} = 8 \text{ V}$$



$$\begin{aligned}
 M1: & R_1 i_{C1} + V_{C0} - E_2 + R_2 i_{E2} = 0 \\
 M2: & -R_2 i_{E2} + E_2 - R_P I_{L0} - N_L = 0 \\
 LKC: & I_{L0} + i_C - i_{E2} = 0
 \end{aligned}
 \Rightarrow \begin{cases} R_1 i_{C1} + V_{C0} - E_2 + R_2 i_{E2} = 0 \\ -R_2 i_{E2} + E_2 - R_P I_{L0} - N_L = 0 \\ I_{L0} + i_C - i_{E2} = 0 \end{cases}$$

$$\Downarrow \\
 i_{E2} = I_{L0} + i_C$$

$$R_1 i_{C1} + V_{C0} - E_2 + R_2 (I_{L0} + i_C) = 0$$

$$\rightarrow i_{C1} (R_1 + R_2) = E_2 - V_{C0} - R_2 I_{L0} \rightarrow i_{C1} = \frac{E_2 - V_{C0} - R_2 I_{L0}}{R_1 + R_2}$$

$$\rightarrow i_{C1} = 0 \text{ A}$$

$$\Rightarrow i_{E2} = I_{L0}$$

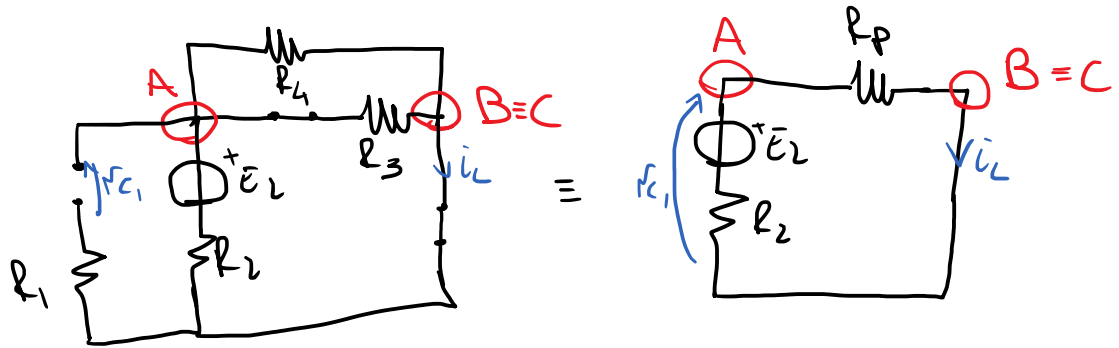
$$\begin{aligned}
 -R_2 I_{L0} + E_2 - R_P I_{L0} - N_L &= 0 \rightarrow N_L = E_2 - (R_2 + R_P) I_{L0} \\
 &= 4 \text{ V}
 \end{aligned}$$

$$i_{C1}(t=0^+) = 0$$

$$\frac{di_L}{dt} \Big|_{t=0^+} = \frac{N_L(t=0^+)}{L} = \frac{4 \text{ V}}{2 \cdot 10^{-3} \text{ H}} = 2 \text{ kA/s}$$

$t = 0$

$t = \infty$

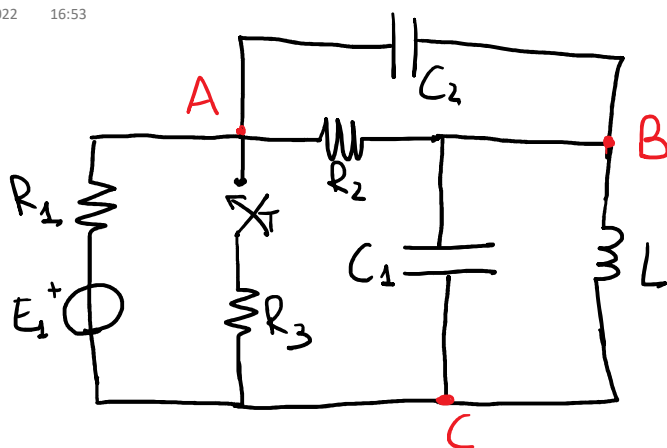


$$i_L(t=\infty) = \frac{E_2}{R_2 + R_p} = 3 \text{ A}$$

$$V_{C1}(t=\infty) = R_p i_L = 6 \text{ V}$$

$$\mathcal{E}_L = \frac{1}{2} L I^2 = \frac{1}{2} L (3)^2 = \frac{1}{2} \cdot 2 \cdot 10^{-3} \cdot 9 = 9 \text{ mJ}$$

$$\mathcal{E}_C = \frac{1}{2} C V^2 = \frac{1}{2} C (6)^2 = \frac{1}{2} \cdot 10 \cdot 10^{-6} \cdot 36 = 180 \text{ nJ}$$



Calcolare: T si chiude $t=0$

$$t=0^-: \mathcal{E}_C = ?$$

$$t=0^+: \frac{di_L}{dt} = ? \quad P_{E_1} = ?$$

$$t=\infty: \mathcal{E}_{C_1} = ? \quad \mathcal{E}_{C_2} = ? \quad \mathcal{E}_L = ?$$

$E_1 = \text{costante} \rightarrow \text{Reg. stazionaria}$

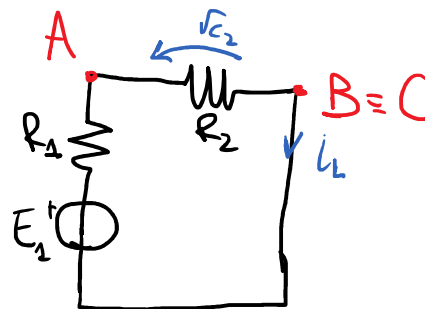
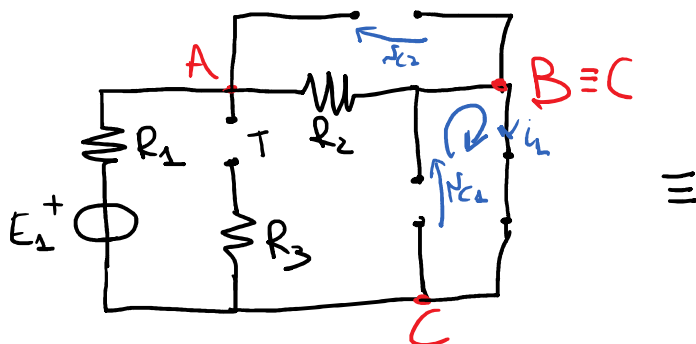
① $t=0^-$

T aperto, regime stazionario $\rightarrow \frac{d}{dt} = 0$



$$i_L = L \frac{di_L}{dt} \xrightarrow{\frac{d}{dt}=0} i_L = 0$$

$$i_C = C \frac{di_C}{dt} \xrightarrow{\frac{d}{dt}=0} i_C = 0$$



Come prima cosa calcoliamo le variabili di stato all'istante $t=0^-$:

$$\mathcal{E}_{C_1}(t=0^-) = 0 \text{ V}$$

$$i_L(t=0^-) = \frac{E_1}{R_1 + R_2}$$

$$\mathcal{E}_{C_2}(t=0^-) = \mathcal{E}_{R_2} = R_2 \cdot i_L = E_1 \cdot \frac{R_2}{R_1 + R_2}$$

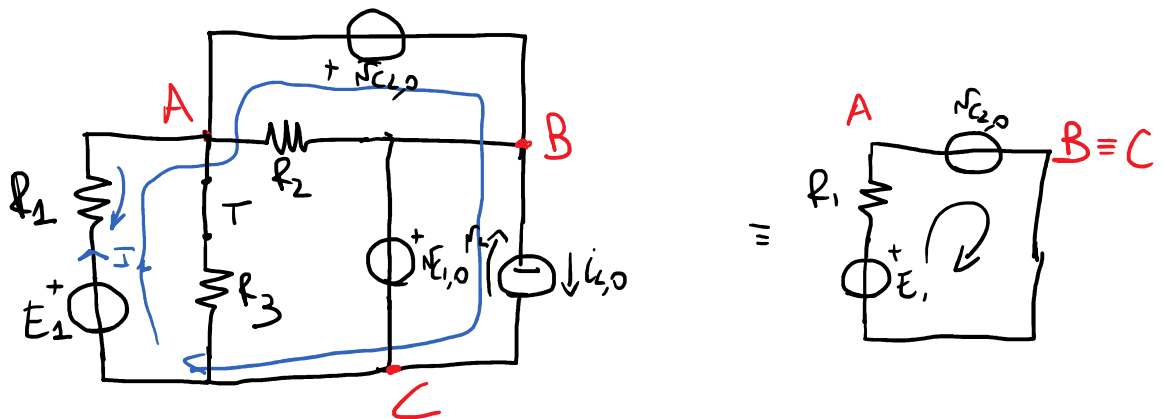
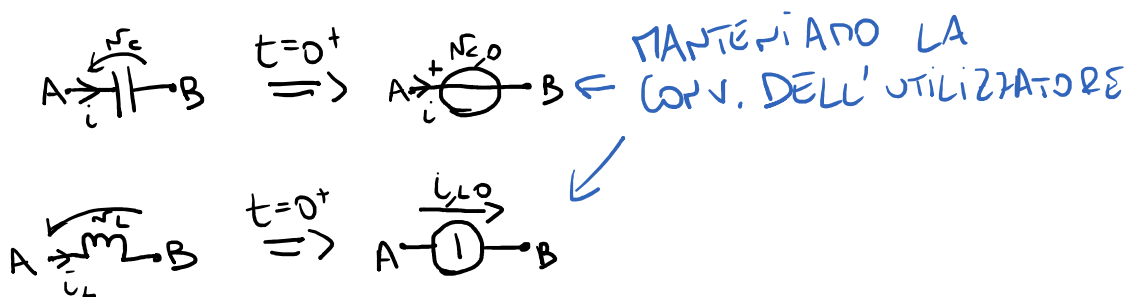
$$\sum C_2 = \frac{1}{2} C V_C^2 = \frac{1}{2} \cdot C \left(E_1 \cdot \frac{R_2}{R_1 + R_2} \right)^2$$

② $t = 0^+$

$$N_{C_1}(t=0^+) = N_{C_1}(t=0^-) = 0 \text{ V}$$

$$N_{C_2}(t=0^+) = N_{C_2}(t=0^-) = N_{C_2,0} = \bar{E}_1 \frac{R_2}{R_1 + R_2}$$

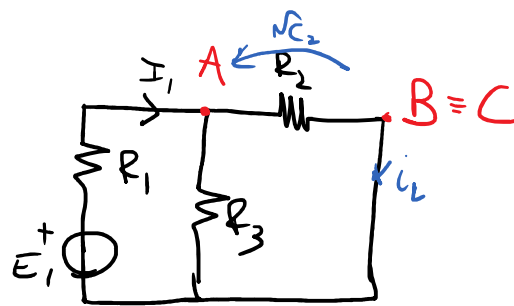
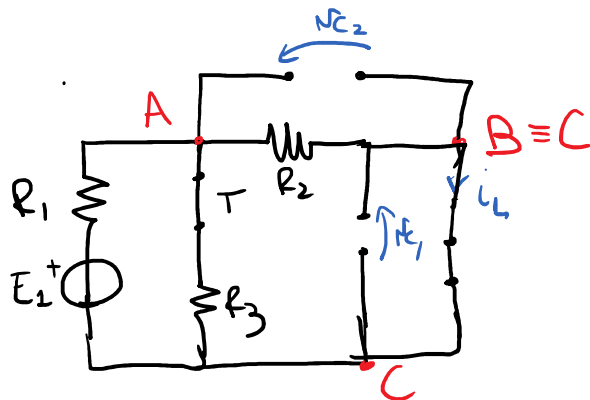
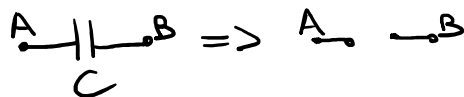
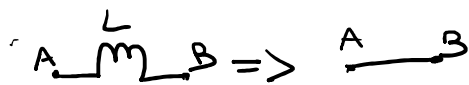
$$i_L(t=0^+) = i_L(t=0^-) = i_{L,0} = \frac{\bar{E}_1}{R_1 + R_2}$$



$$\left. \frac{di_L}{dt} \right|_{t=0^+} = \left. \frac{N_L}{L} \right|_{t=0^+} = \frac{N_{C_1,0}}{L} = 0 \text{ [A/s]}$$

$$P_{E_1} = E_1 \cdot I_1 = E_1 \cdot \left(\frac{\bar{E}_1 - N_{C_2,0}}{R_1} \right) \text{ [W]}$$

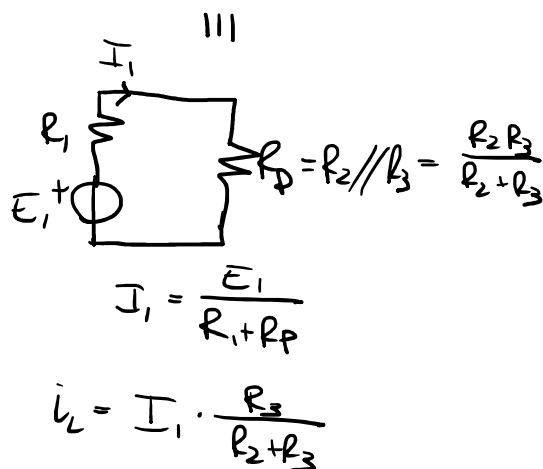
③ $t = \infty \rightarrow$ Reg. stazionario:



$$N_{C_1}(t=\infty) = 0 \text{ V}$$

$$i_L(t=\infty) = \frac{E_1}{R_1 + R_p} \cdot \frac{R_3}{R_2 + R_3}$$

$$N_{C_2}(t=\infty) = R_2 \cdot i_L$$



$$\mathcal{E}_{C_1} = \frac{1}{2} C_1 V_{C_1}^2 = \frac{1}{2} C_1 (0)^2 = 0 \text{ [J]}$$

$$\mathcal{E}_{C_2} = \frac{1}{2} C_2 V_{C_2}^2 = \frac{1}{2} C_2 (R_2 \cdot i_L)^2 \text{ [J]}$$

$$\mathcal{E}_L = \frac{1}{2} L I_L^2 = \frac{1}{2} L \left(\frac{E_1}{R_1 + R_p} \cdot \frac{R_3}{R_2 + R_3} \right)^2 \text{ [J]}$$

La R per la quale ho lo smorzamento critico :

$$\omega = \omega_0$$

$$\downarrow$$
$$\frac{R}{2L} = \frac{1}{\sqrt{LC}} \Rightarrow R = \frac{2L}{\sqrt{LC}}$$